Scalability & Future Plan

Date	07 July 2026
Team ID	SWTID-2026-2168
Project Name	EduGenie: Google Gemini-Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Currently optimized for a limited number of concurrent users.	Response time may increase during heavy usage.	High
2	User data and learning history are not permanently stored in a dedicated database.	Personalized learning experience is limited.	High
3	AI responses depend on external Google Gemini API availability and rate limits.	Temporary delays may occur during high API usage.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for small-scale deployment.	Deploy on cloud platforms with auto-scaling and load balancing.
2	Data Storage	Limited local storage.	Integrate PostgreSQL or MongoDB for secure and scalable data management.
3	Performance	Single FastAPI server instance.	Introduce caching, asynchronous processing, and optimized API handling.
4	Security	Basic API key protection.	Implement user authentication, encrypted communication (HTTPS), and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User Authentication and Personal Profiles	Next Release	Personalized learning experience and secure access.
Phase 2	Learning Progress Tracking and Analytics Dashboard	2–3 Months	Help students monitor academic performance and study consistency.
Phase 3	Multi-language Support and Voice-based Learning	4–6 Months	Improve accessibility for users with different language preferences.
Phase 4	Mobile Application, Cloud Deployment, and Collaboration Features	Future Version	Increase scalability, accessibility, and support for a larger user base.